Subject: Manufacturing Processes (23A0340IT)

UNIT - IV : SHEET METAL FORMING

Blanking & piercing, forces, deep drawing, stretch forming, bending & spring back, coining, spinning, presses & press tools, high energy rate forming (explosive, ~~electromagnetic, electro-hydraulic, rubber pad~~)

1. Introduction to Sheet Metal Forming

Sheet metal working deals with thin plates (thickness 0.4 – 6 mm) shaped on presses using punch and die. Surface-area-to-volume ratio is high; the thickness remains nearly constant (except shearing). Products: car bodies, aircraft skins, utensils, cans, enclosures, appliances, electrical laminations.

- Operations are of three kinds: cutting (shearing) – blanking, piercing, notching, trimming, parting; bending; drawing / forming – deep drawing, stretching, spinning, coining, embossing.
- Sheet materials: low-carbon steel, stainless steel, Al alloys, brass, copper, Ti. Required: high ductility, high n, high plastic strain ratio r (anisotropy).

2. Blanking & Piercing

Both are shearing operations: sheet is sheared between a punch and a die having a small clearance.

- Blanking: punching out a piece (the blank) from the sheet – blank is the product, remaining strip is scrap.
- Piercing: making holes – punched slug is scrap, the sheet with hole is the product.
- Other: notching (cutting from edges), lancing, trimming (removing flash), shaving (fine finishing cut), fine blanking (very smooth edge, triple-action press), parting (separating).

Clearance, forces & power (important formulae)

- Clearance c (per side) = 0.05 to 0.10 t (soft/ thin ~ 3–5 % t; hard ~ 8–10 % t); or c = 0.0032 t sqrt(tau). Blanking: die = blank size, punch smaller by 2c. Piercing: punch = hole size, die larger by 2c.
- Too small clearance: more force, secondary shear, rapid wear. Too large: burrs, rounded edge, rollover.
- Shearing (cutting) force $F = L \times t \times \tau_s$ (L = perimeter of cut, t = thickness,

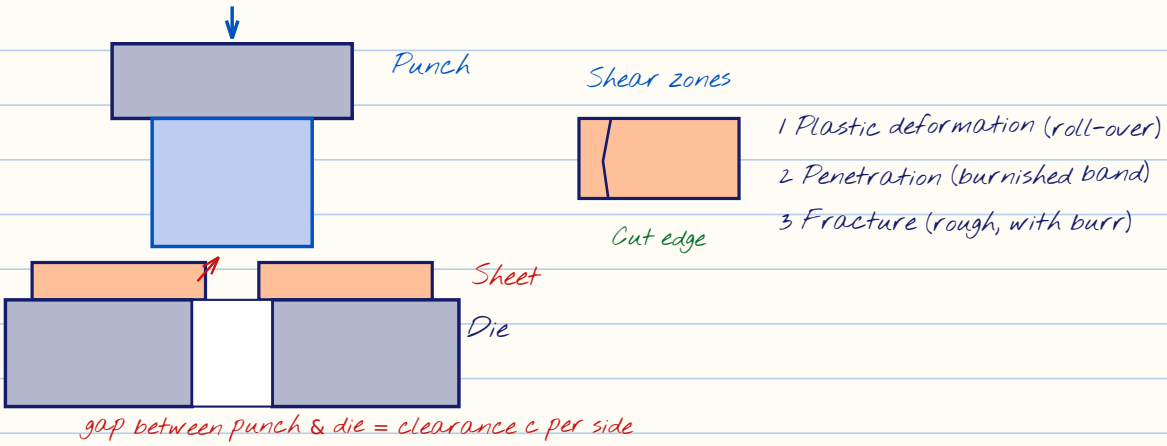


Fig 2.1 Blanking / piercing die set & cut-edge zones

tau_s = shear strength ~ 0.7 x UTS). For round hole: F = pi d t tau.

- Work done W = F x t x K (K = 0.3–0.6 fraction of t for penetration). Power = W x strokes/min.
- Stripping force = 2.5–10 % of cutting force (for hold-down). Press capacity = 1.2 x F.

Reducing cutting force: shear on punch / die

- Grinding a shear (bevel) angle on the punch or die face (shear = about t to 2t) makes cutting progressive - the force falls to 30–60 %, but the stroke increases. For blanking put the shear on the die so the blank stays flat; for piercing put it on the punch so the sheet (product) stays flat.
- Staggering punches (different lengths) in multiple-hole dies also reduces peak load.

Solved example 1 - blanking

Blank 50 mm dia, t = 2 mm, UTS = 400 MPa, tau = 0.7 x 400 = 280 MPa. F = pi x 50 x 2 x 280 = 87.96 KN (~ 8.97 t). With clearance 0.075 t = 0.15 mm/side: die size = 50 mm; punch = 50 - 0.3 = 49.7 mm. Press selected = 1.2 x 88 = about 106 KN.

Sheet-metal cutting operations at a glance

Operation	Description	Product / scrap
Blanking	Cut closed outline from strip	Blank is product
Piercing / punching	Hole made by punch	Slug is scrap
Notching	Removal of metal from edge of sheet	Notched part
Lancing	Cut and bent in one stroke, no metal removed (louvres, tabs)	Part with tab

Slitting / parting	Straight cut separating strip; two cut-offs	Two pieces
Trimming / shaving	Removing flash or excess; shaving gives smooth edge (0.1–0.2 mm)	Finished edge
Fine blanking	V-ring impingement, small clearance (0.5 %); fully burnished edges	Gears, cams

Strip layout & economy

- Strip layout aims to minimise scrap: nest blanks (staggered rows), provide bridge (web) between blanks and edge margin of about 1–1.5 t (min 1.5 mm).
- Material utilisation = (area of blanks per stroke / area of strip fed per stroke) × 100 – target above 70–75 %.
- Scrap often reused for small parts. Die life 10^5 – 10^6 strokes; resharpening by grinding.

3. Bending

Bending deforms the sheet about a straight axis (neutral axis): the outside is in tension, inside in compression; the neutral axis shifts toward the inner side (K factor 0.33–0.5).

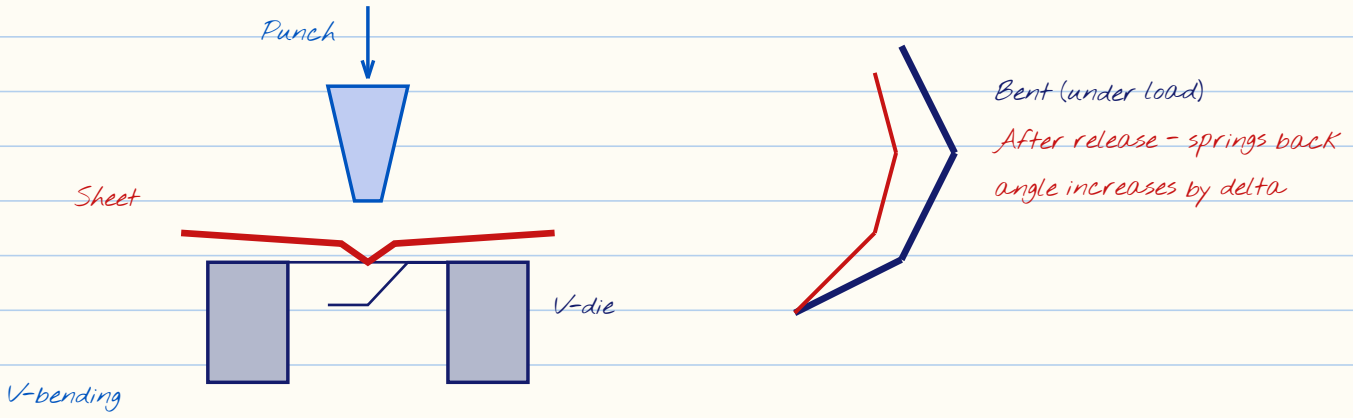


Fig 3.1 V-bending and spring-back

- Types: V-bending, edge (wiping) bending, U-bending, channel, curling, flanging, hemming, seaming, roll bending, air bending, bottoming, coining.
- Bend allowance (length of neutral axis): $BA = \alpha (R + K t)$ (α in radians, R = inner radius, K = 0.33 for $R < 2t$, 0.5 for $R > 2t$). Developed length = sum of straight lengths + BA.
- Minimum bend radius: $R_{min} \sim (0.5 \text{ to } 4)\, t$ – smaller for ductile; bend across rolling direction to avoid cracking.
- Bending force: $F = K_f \, UTS \, L \, t^2 / W$ (L = length of bend, W = die opening ~ 8 t,

K_f = 1.33 for V-die, 0.33 for wiping/ edge).

Spring back & remedies

After removal of load, the elastic recovery makes the bend angle open slightly - spring-back. It increases with higher yield strength, larger R/t, smaller thickness, and large die opening. Ratio $R_i/R_f = 4(R_i \sigma_y / E t)^3 - 3(R_i \sigma_y / E t) + 1$.

- Over-bending: bend more than needed so it springs to the correct angle (compensating angle).
- Bottoming / coining the bend: apply high pressure at end of stroke so that the sheet is plastically set.
- Stretch bending (apply tension while bending), ironing the bend zone, hot / warm bending.
- Use smaller die opening, lower R/t, punch with relief (spring-forming dies), and select low yield-strength / high-E material.

Solved example 3 - bending

Bend a 3 mm thick, 100 mm wide strip through 90 degrees with inner radius 6 mm (R = 2t so K = 0.33-0.5; take 0.4). $BA = (\pi/2)(6 + 0.4 \times 3) = 1.571 \times 7.2 = 11.3$ mm.

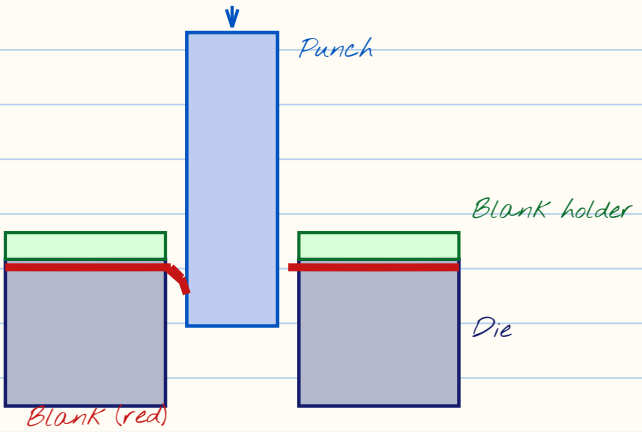
V-die opening W = 8t = 24 mm, UTS = 400 MPa, K_f = 1.33: $F = 1.33 \times 400 \times 100 \times 3^2 / 24 = 19.95$ KN (about 2 t).

Roll forming, tube bending & hydroforming

- Roll forming: strip passes through successive roll stands (6-20) that progressively bend it into long channels, roof sheets, door frames, tubes - continuous, fast.
- Tube bending: rotary draw bending with mandrel to prevent flattening; bend radius >= 1.5 d.
- Hydroforming: fluid pressure (up to 100 MPa) pushes a tube or sheet against a die - complex hollow auto parts (chassis rails).
- Superplastic forming: Ti / Al sheet at 900/1500 C stretched with gas pressure into a die (elongation > 300 %).

4. Deep Drawing

Deep drawing converts a flat blank into a hollow cup / box by pulling it into a die cavity with a punch; the flange is drawn inward (radial tension, circumferential compression). A blank holder presses the flange to prevent wrinkling. Examples: Kitchen sinks, pans, cans, shells, fuel tank halves.



DR = D / d (draw ratio); limiting DR ~ 2

First draw reduction 40–50 %

Radii: die = 4–10 t, punch = 4–10 t

Clearance = 1.1 – 1.2 t

Blank dia: $D = \sqrt{(d^2 + 4 d h)}$

Draw force $F = \pi d t \cdot UTS (D/d - 0.7)$

Fig 4.1 Deep drawing of a cup

- Blank diameter for a cylindrical cup of diameter d & height h (area equality): $D = \sqrt{(d^2 + 4 d h)}$.
- Drawing ratio / limiting draw ratio (LDR) = D/d_{max} ~ 1.8 – 2.2. Reduction r = (D – d)/D (max 0.5). Deeper cups need redrawing (reverse / direct, with inter-stage annealing); ratio of later draws 1.3–1.2.
- Drawing force: $F = \pi d t \cdot UTS (D/d - 0.7)$ approx; blank holding force = 1/3 of drawing force (pressure 1–3 MPa).
- Lubrication: oil, wax, polythene film, soap (reduces friction), die & punch polished.
- Ironing: further thinning of wall by passing through a clearance < t (cans). Drawing in double-action press: outer ram = blank holder, inner ram = punch.

Defect	Cause	Remedy
Wrinkling (flange / wall)	Low blank-holder force, thin sheet, large flange, die radius large	Increase holding force, draw beads, thicker sheet
Tearing / fracture at cup bottom	Too high draw ratio, sharp radii, high friction, low holder force	Increase radii, lubricate, anneal, reduce DR, redraw
Earing	Planar anisotropy of rolled sheet (variation in r)	Use sheet with low delta-r; trim ears
Surface scratches / orange peel	Dirty dies, coarse grain	Polish dies, fine-grain sheet

Stretcher strains (Luders bands)	Yield-point elongation in low-C steel	Temper-roll (skin pass) the sheet

Solved example 2 - deep drawing

Cup $d = 50$ mm, $h = 40$ mm. Blank $D = \sqrt{(50^2 + 4 \times 50 \times 40)} = \sqrt{(2500 + 8000)} = 102.5$ mm; $DR = 102.5/50 = 2.05$ (> 2): two draws are needed. First draw to $d1 = 102.5/1.8 \sim 57$ mm, second to 50 mm.

5. Stretch Forming, Coining & Spinning

Stretch forming

Sheet is gripped at its edges by jaws, stretched beyond yield (tension) and wrapped over a form block (die). Because of nearly uniform tension, spring-back & wrinkling are small. Used for large, shallow contoured panels - aircraft skins, wing leading edges, car roof panels, curved architectural panels.

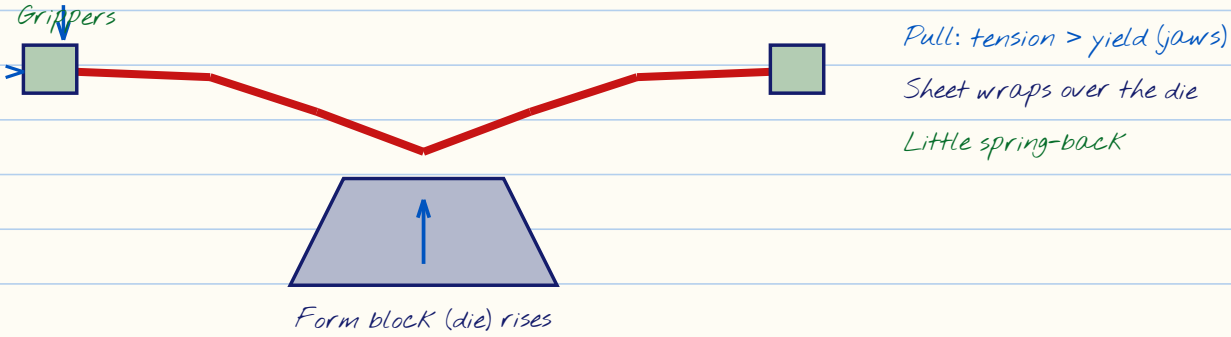


Fig 5.1 Stretch forming

Coining & embossing

- Coining: sheet / blank squeezed between closed dies under very high pressure (up to 5-6 times flow stress, 1600 MPa); metal flows to fill the finest details on both faces - no flash; thickness changes. Coins, medals, jewellery, hallmark, spring-back removal (bottoming).
- Embossing: shallow relief in a sheet by matching punch & die - thickness nearly the same (name plates, ribs). Coining force = $p \times A$ (p = 3-5 x UTS).

Spinning

A flat disc / tube clamped against a mandrel (former) on a lathe is rotated; a roller or tool progressively forces the metal over the mandrel. Products are axisymmetric: cones, hemispheres, reflectors, cooking vessels, rocket nose cones.

- Conventional (manual) spinning: wall thickness stays constant, diameter reduces - small quantities, cheap tooling.
- Shear spinning (power spinning): roller thins the wall: $t_f = t_0 \sin(\alpha)$ (sine law), diameter constant; accurate cones, no blank wrinkling. Tube spinning: reduces tube wall.
- Merits: low tooling cost, large parts, good finish, strain hardening. Limits: only symmetrical shapes, skilled labour.

6. Presses & Press Tools

A press is a machine tool with a ram (slide) reciprocating towards a fixed bolster - supplies force and energy for sheet operations. Capacity is in tonnes.

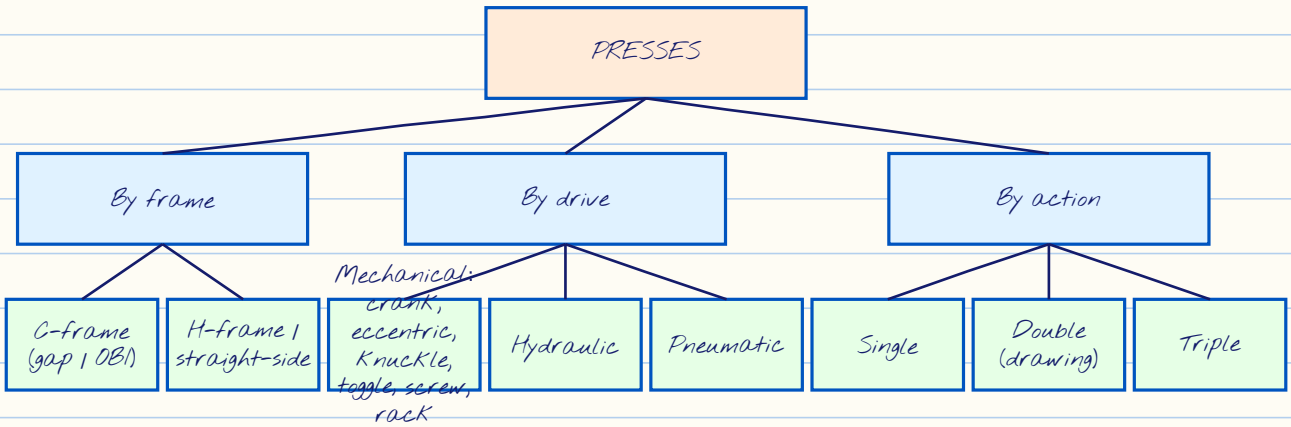


Fig 6.1 Classification of presses

Press	Characteristics & use
Mechanical (crank / eccentric)	Flywheel energy through clutch; fixed stroke; fast (20-1000 spm); blanking, piercing, shallow forming; force greatest at bottom dead centre.
Hydraulic	Oil pressure on piston; full force at any stroke, adjustable, slow - deep drawing, forming, forging, large parts; overload-safe.
Knuckle-joint / toggle	Huge force near end of stroke with slow ram speed: coining, embossing, sizing.
Screw press	Energy stored in flywheel & screw; used in forging, coining, minting.
Single / double / triple action	Double has blank-holder ram + punch ram (deep drawing); triple adds a bottom ram (reverse draw, ejector).

C-frame vs H-frame	C-frame - open front, easy access, lower rigidity (small work); H / straight-side - rigid, for heavy jobs & accuracy.

Press tools (dies)

- Simple (single-operation) die: one operation per stroke; low cost; low volume.
- Compound die: two or more cutting operations at one station in one stroke - e.g. blanking + piercing a washer; accurate concentricity, slow strip, parts fall back through die.
- Progressive die: strip moves through several stations; each stroke performs a different operation; finished part is cut off at the last station - high production, automatic.
- Combination die: cutting + non-cutting (blank + draw) in one stroke. Transfer die: separate blank moved by fingers between stations (large parts).

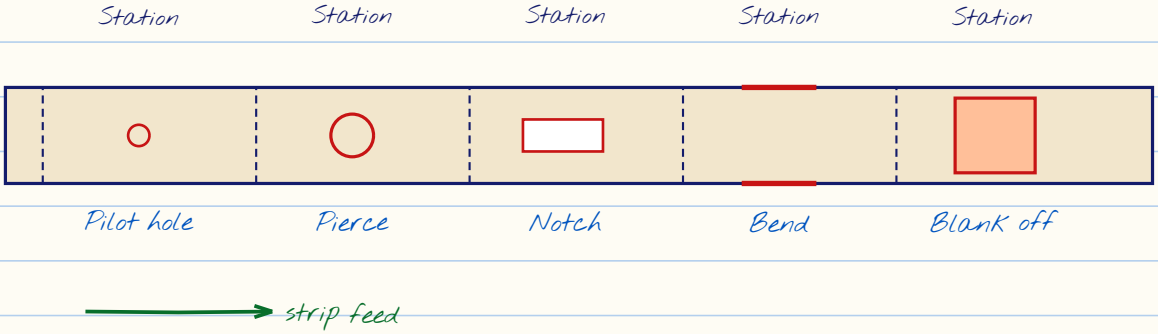


Fig 6.2 Progressive die - strip layout

Components of a press tool

- Punch & die, punch plate (holder), die block / shoe, stripper plate (strips sheet off the punch), guide pillars & bushes, pilot (locates strip), stops / gauges, shank, back-up plate, knock-out, blank-holder (draw dies), springs, fasteners & dowels.
- Materials: tool steel D2 / D3 (12 % Cr), A2, oil-hardening O1, high-carbon steel, tungsten carbide inserts; hardness 58-62 HRC. Die life improves with lubrication & correct clearance.

Formability of sheet metal

- Strain-hardening exponent n: high n -> better stretchability (uniform elongation). Plastic strain ratio r = width strain / thickness strain: high r (> 1) -> good deep drawing (resists thinning); average $r = (r0 + 2\,r45 + r90)/4$;

planar anisotropy delta-r = (r0 - 2 r45 + r90)/2 -> earing.

- Forming limit diagram (FLD): plot of major vs minor strain at the onset of necking - safe region below the curve.
- Tests: Erichsen cupping, Olsen, Swift cup, hydraulic bulge, limiting dome height (LDH).

Formula sheet - Unit IV

Item	Formula
Shearing force	F = L t tau (tau ~ 0.7 UTS); round: pi d t tau
Work / power	W = F x K t ; P = W x strokes per min
Clearance (per side)	c = 0.05 - 0.1 t
Bend allowance	BA = alpha (R + K t)
Bending force (V)	F = 1.33 UTS L t^2 / W
Blank diameter (cup)	D = sqrt(d^2 + 4 d h)
Drawing force	F = pi d t UTS (D/d - 0.7)
Draw ratio / reduction	DR = D/d (<= 2); r = 1 - d/D (<= 0.5)

Quick viva

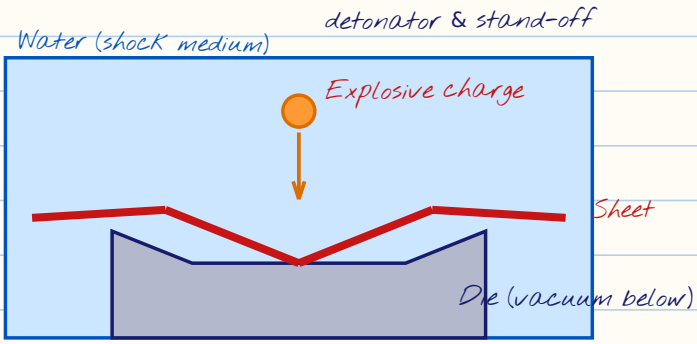
- Why shear on punch? -> To lower the peak cutting force. Why blank holder? -> To prevent wrinkling of flange.
- Which press for deep drawing? -> Double-acting / hydraulic press. Why coining needs a toggle press? -> Very high pressure at the end of stroke.
- What is spring-back? -> Elastic recovery of the bend after load is removed.

7. High Energy Rate Forming (HERF)

HERF processes release a large amount of energy in a very short time (milliseconds) to deform the metal at very high velocity (30-300 m/s). Useful for large, thin, hard-to-form materials (Ti, Ni alloys, stainless steel) with small production runs, where press cost would be high.

7.1 Explosive forming

Explosive charge (TNT, RDX, PETN, gun powder) is detonated in a medium (usually water) near the sheet; the shock wave / pressure pulse (up to 25 GPa) drives the sheet into the die. The die may be cheap (concrete, cast Zn/ Kirksite) with vacuum to pull the sheet.



Stand-off distance controls pressure

Large parts: ship plates, domes, tanks

Few parts, thick / large, low die cost

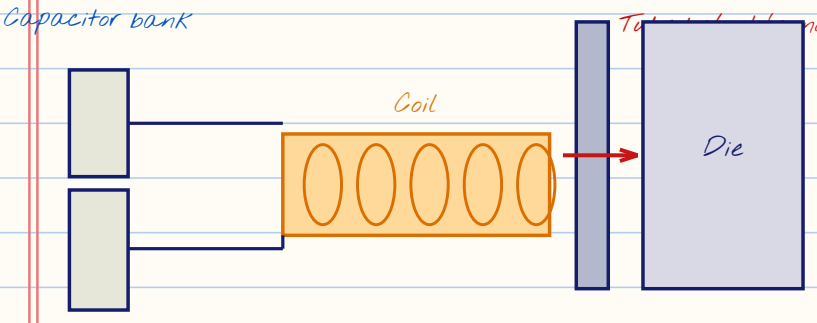
Safety & noise limits

Fig 7.1 Explosive forming (stand-off method)

- Stand-off: charge placed at distance in a medium (water) - uniform loading on the sheet. Contact: charge in contact with the work (for sizing/ cladding/ hardening). Confined: in a closed die; unconfined: open.
- Advantages: no costly press, very large parts, complex shapes, Ti & high-strength alloys, low tooling cost, less spring-back. Limitations: hazardous, skilled safe handling, low production rate, noise, licensing.

7.2 Electromagnetic forming (magnetoforming)

A capacitor bank is discharged through a coil placed near an electrically conductive work-piece; the rapidly changing magnetic field induces eddy currents in the work, which then interacts with the field (Lorentz force) and repels / pushes the workpiece at high speed against the die. Process time ~ 10-100 microseconds.



Used for: tube swaging, bulging, flanging, shallow drawing, assembly

Only conductive metals (Al, Cu, mild steel)

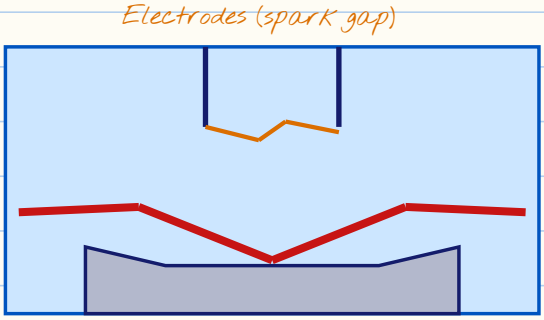
No contact tool, fast, repeatable

Fig 7.2 Electromagnetic (coil) forming

- Merits: contactless - no tool marks, high rate, good repeatability, no lubrication, less spring-back, can join dissimilar tubes (magnetic pulse welding). Limits: conductive materials only, coil life, work shape limited (tubular / shallow), small sheets, expensive power unit.

7.3 Electro-hydraulic forming (EHF)

Electric spark discharge (capacitors, 10-40 KV, 10-100 KJ) between two electrodes under water vaporises the fluid producing a high-pressure shock wave that forms the sheet into the die. Similar to explosive forming but easier to control and safe.

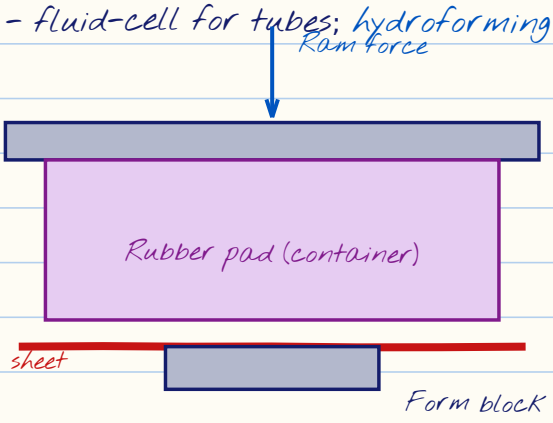


Spark vaporises water -> shock wave
No explosives; safer than explosive forming
Lower energy -> smaller parts, tubes, domes
Slower than EM; die vacuum used

Fig 7.3 Electro-hydraulic forming

7.4 Rubber pad forming

A rubber (polyurethane) pad replaces one half of the die. The sheet is pressed by the ram against a single form block; rubber behaves like a fluid (hydrostatic) pressing the sheet into the shape. Guerin process - rubber pad in a container, force up to 10 MPa+; Marform - rubber + blank holder for deep parts; Verson-Wheelon - fluid-cell for tubes; hydroforming - flexible diaphragm with fluid pressure.



Only one rigid die needed (cheap)
Shallow parts, brackets, aircraft ribs
No tool marks; many parts at once
Low pressure - small depth; pad wears

Fig 7.4 Rubber pad (Guerin) forming

Process	Energy source	Typical use	Advantage	Limitation
Explosive	Chemical explosive	Large domes, tanks, ship / aerospace	Huge parts, cheap die	Hazard, slow cycle
Electro-magnetic	Capacitor + coil	Tubes, shallow forms, assembly	Contactless, fast	Conductive only
Electro-hydraulic	Spark in water	Small/medium parts, tubes	Controlled, safe	Low energy, electrode wear

Rubber pad	Press + elastomer	Shallow sheet parts	Low tool cost	Limited depth, pad wear

General advantages & limitations of HERF

- Advantages: form difficult metals, less spring-back, little or no diel press cost, large parts, good accuracy & finish, strain-rate raises formability.
- Limitations: safety & training, noise, low production rate, limited shapes, energy control needed, special facilities (pits, tanks).

Important exam questions - Unit IV

- Differentiate blanking & piercing; clearance; calculate cutting force & power. (10 M)
- Deep drawing: explain with sketch; draw ratio; defects. Stretch forming & spinning (5 M).
- Bending: bend allowance, spring-back & remedies. Coining (5 M).
- Types of presses & press tools (compound, progressive...). (10 M)
- Explain explosive / electromagnetic / electro-hydraulic / rubber pad forming; advantages & limitations. (10 M each)